Raul Garcia-Lopez

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PROFESSIONAL SUMMARY

Computer Engineering graduate (Cal Poly Pomona, Dec 2025) with hands-on experience designing and deploying high-availability Linux infrastructure with automated failover, real-time monitoring, and centralized logging, plus AWS-based security and data-pipeline work. Strong foundation in TCP/IP networking, operating systems, and systems programming. Seeking an entry-level Cloud Engineer role.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EC2, VPC), VMware, Keepalived (VRRP), Virtual IP failover, Nginx reverse proxy
Monitoring & Observability: Prometheus, Grafana, Alertmanager, Node Exporter
Logging & Data: ELK Stack (Elasticsearch, Logstash, Kibana), OpenSearch, Filebeat, GeoIP enrichment
OS & Systems: Linux (Ubuntu), SSH, systemd, Bash scripting, rsync, cron
Containers & DevOps: Docker (image building & container management), Git
Networking: TCP/IP, reverse proxy architecture, VRRP, network monitoring
Programming: Python (Flask), C, C++, SystemVerilog / Verilog HDL

PROJECTS

Fault-Tolerant Cloud Infrastructure with High Availability — Senior Design Project *Fall 2024*
Linux • Keepalived (VRRP) • Nginx • Prometheus • Grafana • Alertmanager • ELK Stack • Python Flask • rsync • VMware

- Designed and deployed a 3-VM Linux infrastructure simulating a production cloud environment with automated failover and zero-downtime service continuity.
- Configured Keepalived (VRRP) for automatic Virtual IP failover between primary and backup nodes, with Nginx as a reverse proxy that auto-switches routing to the active node on failure.
- Built a Prometheus + Node Exporter monitoring stack and Grafana dashboards (10+ panels) visualizing CPU, memory, disk, network, and live VIP-host tracking across all VMs.
- Integrated Alertmanager webhook automation: alerts trigger a Python Flask listener that rebuilds and restarts the application on the backup VM; a second alert initiates graceful failover when CPU exceeds 95% for 60s.
- Deployed centralized logging (ELK Stack + Filebeat) with rsync/cron directory mirroring between nodes, then stress-tested under simulated failure to confirm sub-minute failover and automatic recovery.

AWS SSH Honeypot — Threat Intelligence Pipeline *2025*
AWS EC2 • Custom VPC • Cowrie • Logstash • OpenSearch • GeoIP • Linux

- Deployed a hardened, internet-facing SSH honeypot (Cowrie) on an isolated AWS EC2 instance — custom VPC, restricted security groups, locked-down egress — capturing 205,620 real attack events over 7 days.
- Built a Logstash → OpenSearch pipeline on a separate, isolated processing tier (mirroring SOC design that keeps tooling off the compromised host), with GeoIP enrichment powering a geographic attack-map dashboard.
- Attributed ~94% of traffic to two automated botnet campaigns and reverse-engineered a post-shell attack playbook (execution check → hardware reconnaissance) from captured command data.
- Validated ingest counts against source data to catch a 4x duplication bug, and corrected OpenSearch geo_point mapping so coordinates rendered on the map — practicing data integrity before trusting visualizations.

ASCON Lightweight Cryptography — Cross-Architecture Benchmarking *Spring 2024*
C • ESP32 • Arduino MEGA • Microblaze FPGA (Nexys A7) • Vitis IDE • UART • Power Analysis

- Implemented the NIST-standardized ASCON encryption algorithm across three architectures: ESP32 (240 MHz), Arduino MEGA 2560 (16 MHz), and a Microblaze soft-core on a Nexys A7-100T FPGA (100 MHz).
- Benchmarked execution time, power, and cost: ESP32 hit a 103x speedup over the MEGA at only 1.14x the power draw (~$9 vs. ~$350 for the Nexys), demonstrating strong performance-per-watt for edge deployments.
- Adapted the C implementation to 32-bit for Microblaze compatibility and recorded FPGA resource utilization (flip-flops, LUTs, BRAM) via Vivado.

Embedded Systems & Digital Design — Coursework *2023–2024*
SystemVerilog • C++ • Verilog HDL • Xilinx FPGA (Nexys A7) • Vivado

- Designed custom FPGA IP cores in SystemVerilog/Verilog (parameterized barrel shifter, asymmetric FIFO, BRAM/ROM lookup modules, VGA display core) and paired them with C++ software drivers in hardware-software co-design projects.
- Built FSM-driven systems (UART lap-timer, Morse-to-ASCII decoder, vending-machine controller) verified through Vivado testbench simulation, and interfaced peripherals including a Pmod time-of-flight distance sensor.

EDUCATION

B.S. Computer Engineering — California State Polytechnic University, Pomona *Graduated Dec 2025*
Relevant Coursework: TCP/IP Internetworking & Lab • Operating Systems for Embedded Applications • Computer Architecture • Data Structures & Algorithms • Advanced Digital Design (Verilog HDL) & Lab